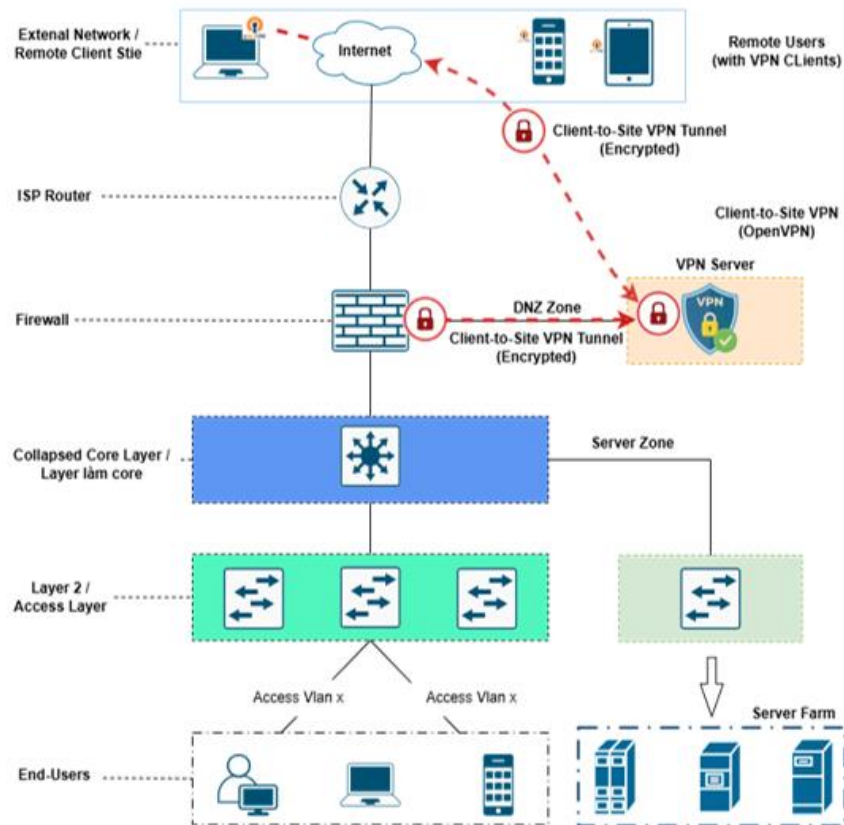
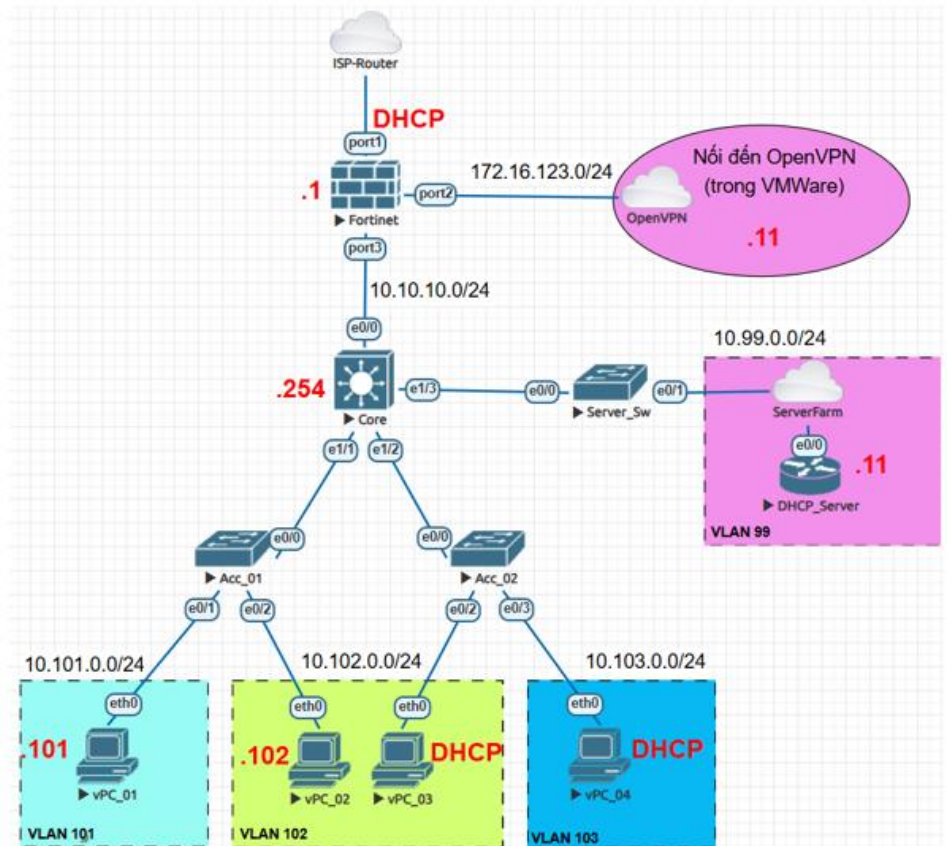


HƯỚNG DẪN THỰC HÀNH LAB

Sơ Đồ Tổng Quan

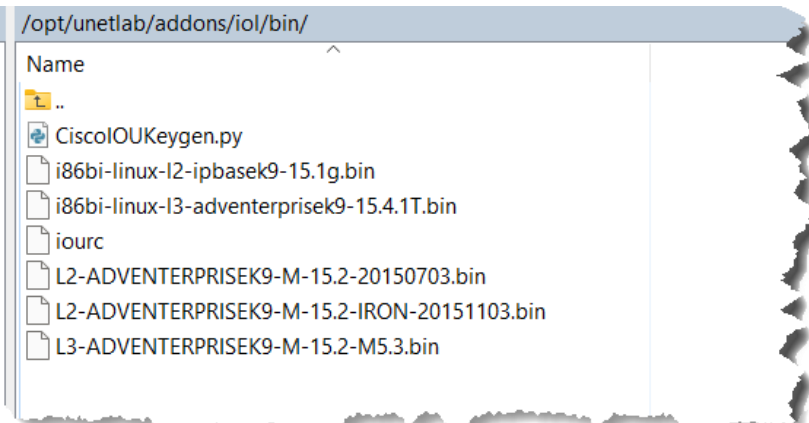
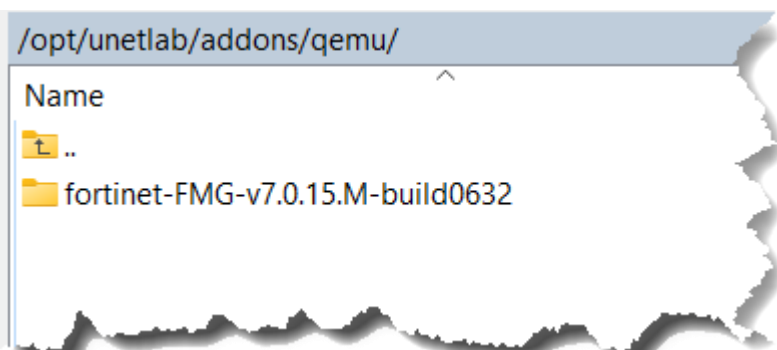


LAB VMware



1. Chuẩn bị môi trường

❖ Import OVA cho EVE-NG

Sau khi import đảm bảo phải có các file image	
- Các file image CISCO sau khi import, nếu không có phải thêm vào - Cách thêm xem hướng dẫn trong source	
- Các file image Fortinet sau khi import, nếu không có phải thêm vào - Cách thêm xem hướng dẫn trong source	

❖ Import sơ đồ lab

Import file sơ đồ từ mạng	
Nếu thành công chọn open	
Hoặc cũng có thể copy file <code>.unl</code> vào <code>/opt/unetlab/labs/</code> và dùng lệnh <code>/opt/unetlab/wrappers/unl_wrapper -a fixpermissions</code>	

❖ Import OVA cho OpenVPN

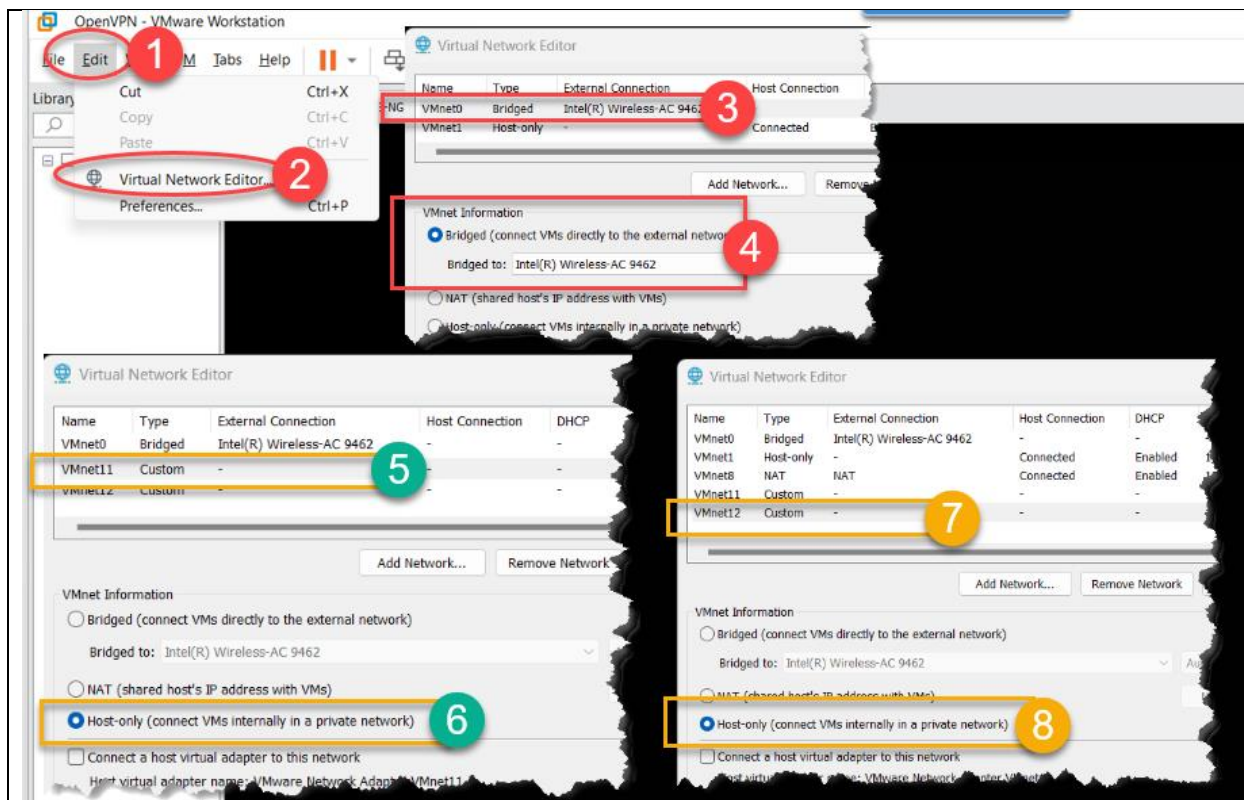
Trong Source demo

❖ OpenVPN Client

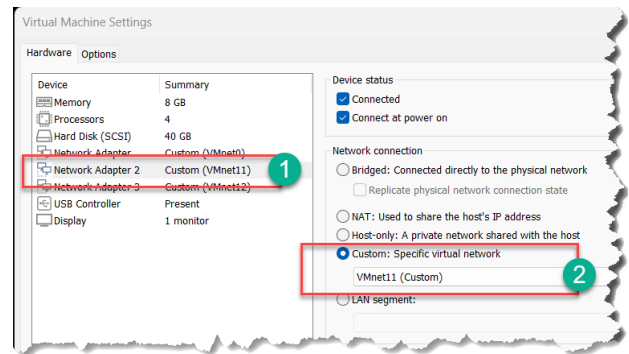
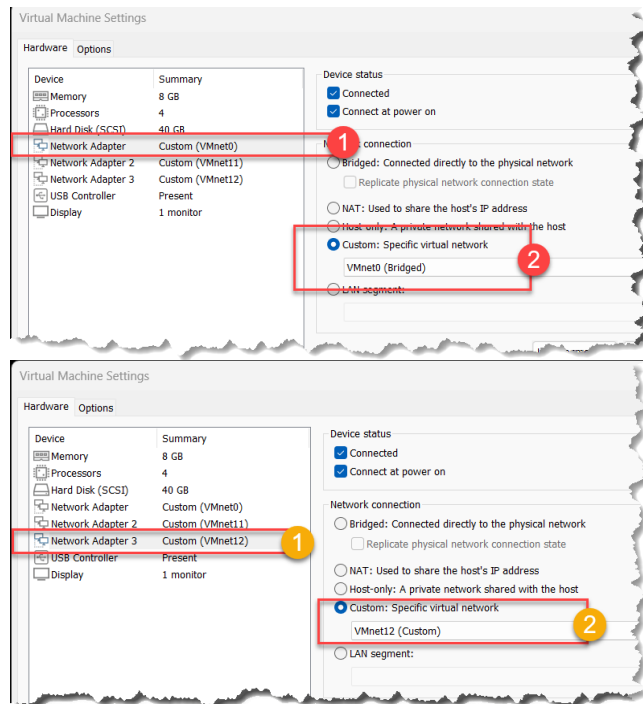
<https://openvpn.net/client/>

2. Trên VMware:

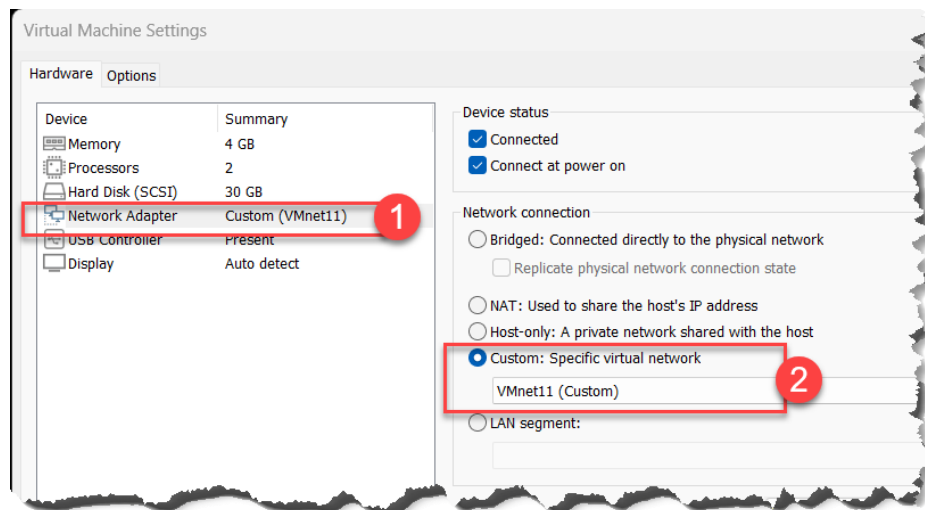
Name	Type	Vmnet Information
Vmnet0	Bridged	Chọn card wireless của máy tính
Vmnet11	Custom	Host-only (Connect VMs internally in a private network)
Vmnet12	Custom	Host-only (Connect VMs internally in a private network)



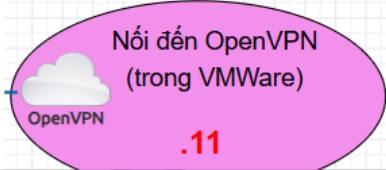
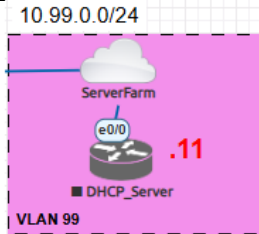
3. Máy ảo EVE-NG



4. Máy ảo OpenVPN



5. Lab EVE-NG

Cloud OpenVPN	Cloud ServerFarm
 Nối đến OpenVPN (trong VMWare) OpenVPN .11 EDIT NETWORK ID: 11 Name/Prefix: OpenVPN Type: Cloud1 Left: 591 Top: 141	 10.99.0.0/24 ServerFarm e0/0 .11 DHCP_Server VLAN 99 EDIT NETWORK ID: 2 Name/Prefix: ServerFarm Type: Cloud2 Left: 717 Top: 303
Cloud ISP Router	
 ISP-Router EDIT NETWORK ID: 1 Name/Prefix: ISP-Router Type: Management(Cloud0) Left: 348	

6. Các Image cho các thiết bị

Link full profile của Firewall fortinet

<https://drive.google.com/file/d/1Lp1a2jEz3NDyZaFOLvN6bHcoTkTAIx8V/view>

Firewall

Template

Fortinet FortiGate

ID

9

Image

fortinet-FMG-v7.0.15.M-build0632

Name/prefix

Fortinet

Icon

Firewall.png

Core Switch/Layer 3

i86bi-linux-l2-ipbasek9-15.1g.bin

Template

Cisco IOL

ID

8

Image

i86bi-linux-l2-ipbasek9-15.1g.bin

Name/prefix

Core

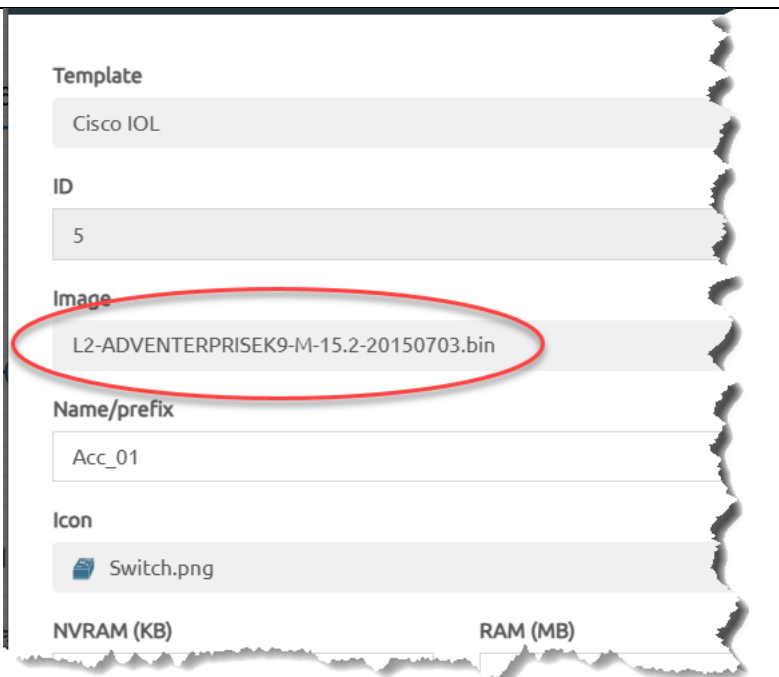
Icon

Switch L3.png

NVRAM (KB)

RAM (MB)

Access Switch/Layer 2
L2-ADVENTERPRISEK9-M-
15.2-20150703.bin



Template
Cisco IOL

ID
5

Image
L2-ADVENTERPRISEK9-M-15.2-20150703.bin

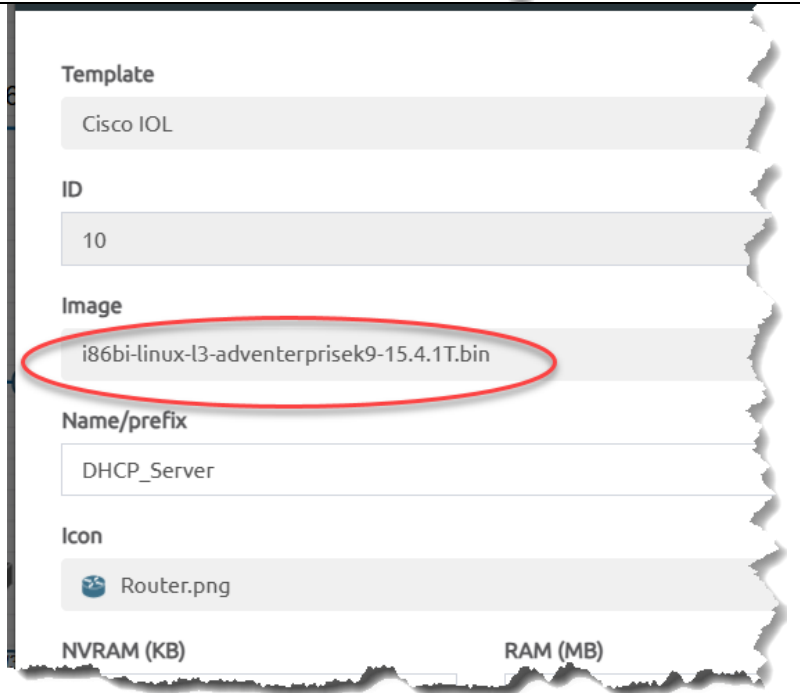
Name/prefix
Acc_01

Icon
Switch.png

NVRAM (KB)

RAM (MB)

Router
L3-ADVENTERPRISEK9-M-
15.2-M5.3.bin



Template
Cisco IOL

ID
10

Image
i86bi-linux-l3-adventerprisek9-15.4.1T.bin

Name/prefix
DHCP_Server

Icon
Router.png

NVRAM (KB)

RAM (MB)

7. Bảng Quy Hoạch Interface, VLAN và Địa Chỉ IP

STT	Tên Thiết Bị	Interface	IP Address	Gateway /Route	Mode	Ghi Chú
1	ISP-Router					Kết nối với mạng LAN/WiFi
2	Firewall	Port 1	DHCP		Routed	WAN
		Port 2	172.16.123.1/24		Routed	DMZ
		Port 3	10.10.10.1/24		Routed	LAN
3	Core	E0/0	10.10.10.2/24	10.10.10.1	Routed	Uplink to FW
		E1/1			Trunk	Acc 1
		E1/2			Trunk	Acc 2
		E1/3			Access	Server Sw
		Vlan 99	10.99.0.254/24			Nội Server Farm
		Vlan 101	10.101.0.254/24			Nhân sự
		Vlan 102	10.102.0.254/24			Kế toán
		Vlan 103	10.103.0.254/24			Kinh Doanh
4	Acc_01	E0/0			Trunk	
		E0/1			Access	Vlan 101
		E0/2			Access	Vlan 102
5	Acc_02	E0/0			Trunk	
		E0/1			Access	Vlan 102
		E0/2			Access	Vlan 102
6	Server_Sw	E0/0			Access	Không cấu hình
		E0/3			Access	
7	vPC 01	Eth0	10.101.0.11/24	10.101.0.254		Vlan 101
8	vPC 02	Eth0	10.102.0.12/24	10.102.0.254		Vlan 102
9	vPC 03	Eth0	DHCP			Vlan 102
10	vPC 04	Eth0	DHCP			Vlan 103
11	DHCP_Server		10.99.0.11/24	10.99.0.254	Routed	Windows Server/Router
12	Open VPN		172.16.123.11/24	172.16.123.1/24		

8. Cấu hình thiết bị

❖ Truy cập vào EVE-NG

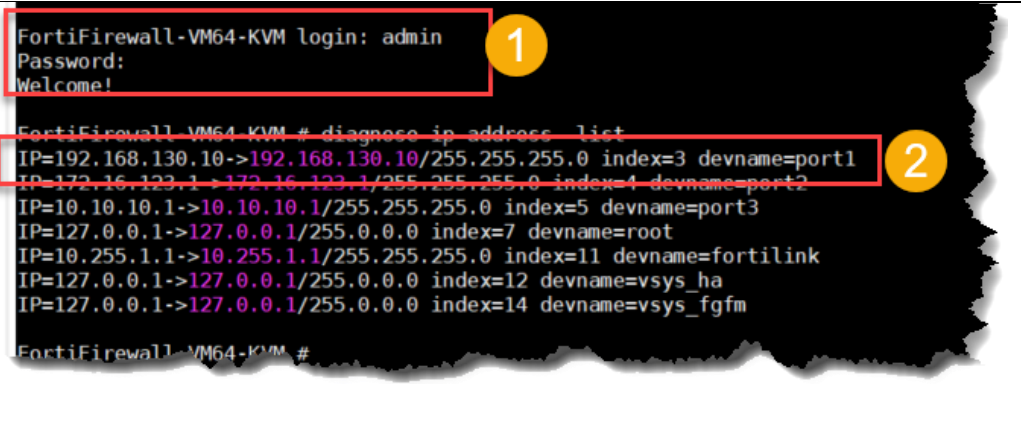
- Xem IP
`Ip -c a | grep inet`
Hoặc
`Ip -c a | grep pnet0`
- Truy cập vào web
Ví dụ:
<http://192.168.130.2>
- ID/PASS: root/eve



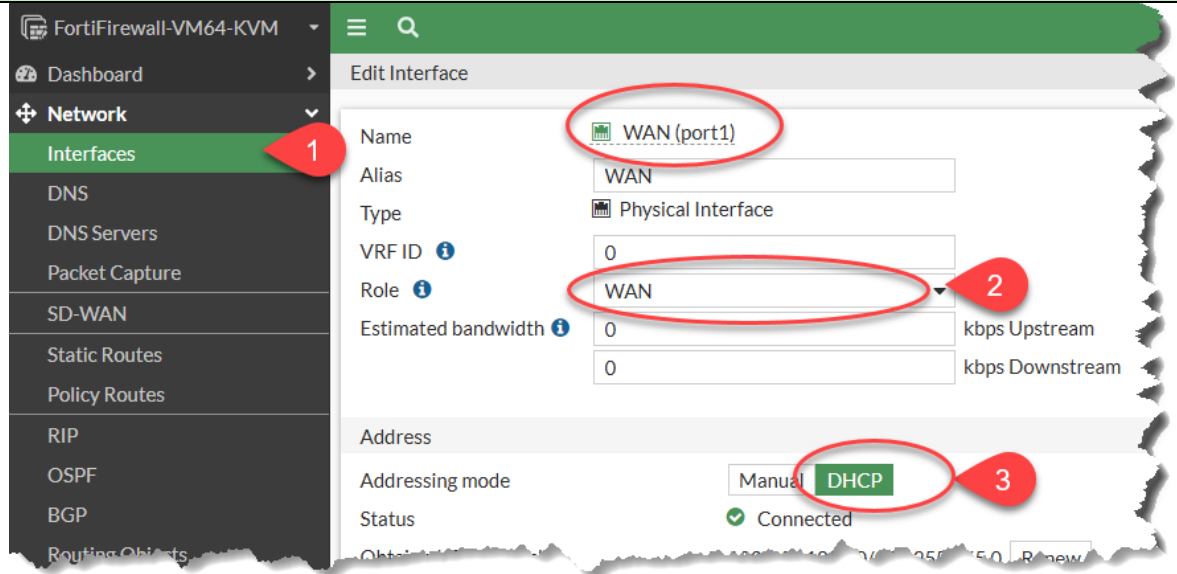
❖ Firewall

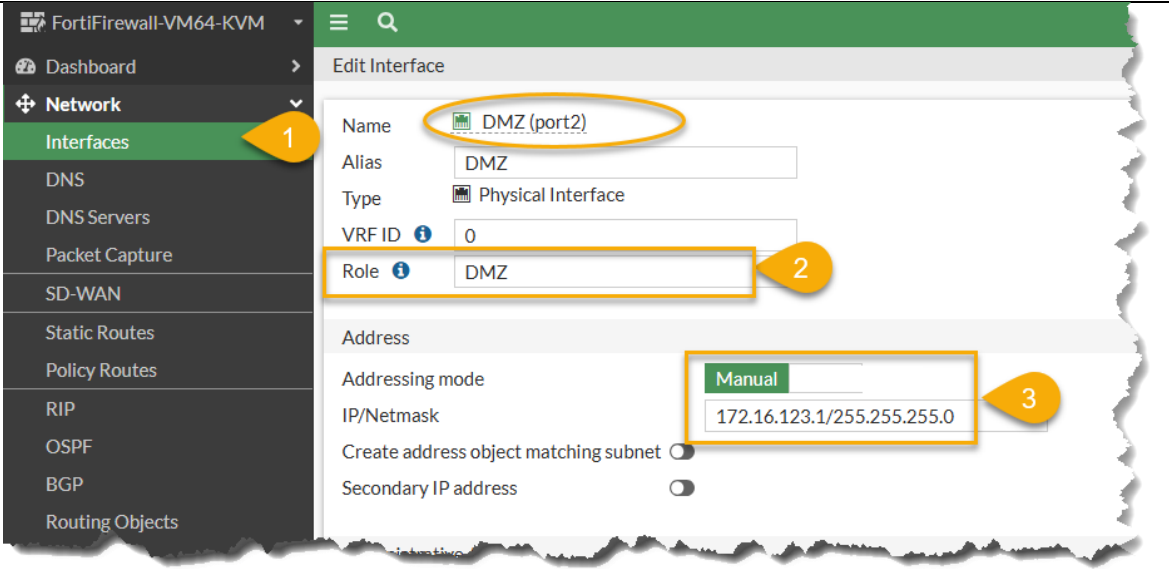
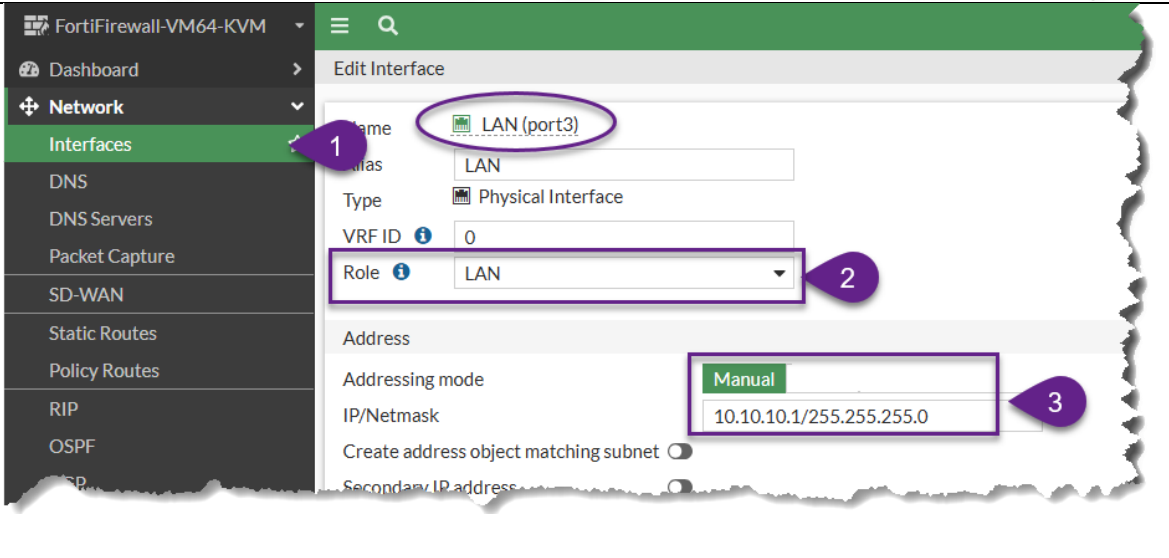
Truy cập vào thiết bị

- **Lệnh kiểm tra ip:**
`diagnose ip address list`
- Là ip trên **port1**
-
- Truy cập vào web với IP trên port 1
Ví dụ:
<http://192.168.130.10>
- ID/PASS: admin/admin



❖ Cấu hình interface

Ghi chú	CLI	GUI
Cấu hình Interface		
Port1 WAN	<pre> config system interface edit "port1" set vdom "root" set mode dhcp set allowaccess ping https ssh fgfm set type physical set alias "WAN" set lldp-reception enable set role wan set snmp-index 1 next end </pre>	 The screenshot shows the FortiGate GUI for configuring the WAN interface. The left sidebar shows the 'Network' menu with 'Interfaces' selected (marked with a red circle and '1'). The main panel shows the 'Edit Interface' configuration for 'WAN (port1)' (marked with a red circle). The 'Alias' is set to 'WAN'. The 'Type' is 'Physical Interface'. The 'Role' is set to 'WAN' (marked with a red circle and '2'). The 'Addressing mode' is set to 'DHCP' (marked with a red circle and '3'). The status is 'Connected'.

Port2 DMZ	<pre> config system interface edit "port2" set vdom "root" set ip 172.16.123.1 255.255.255.0 set allowaccess ping https ssh fgfm set type physical set alias "DMZ" set role dmz set snmp-index 2 next end </pre>	
Port3 LAN	<pre> config system interface edit "port3" set vdom "root" set ip 10.10.10.1 255.255.255.0 set allowaccess ping https ssh fgfm set type physical set alias "LAN" set device-identification enable set lldp-transmission enable set role lan set snmp-index 3 next end </pre>	

Cấu hình Rouing

Cấu hình routig – GUI giải thích

The screenshot displays the FortiGate GUI for configuring static routes. The left sidebar shows the 'Network' menu with 'Static Routes' highlighted (1). The main table lists the configured routes:

Destination	Gateway IP	Interface	Status
10.101.0.0/24	10.10.10.2	LAN (port3)	Enabled
10.102.0.0/24	10.10.10.2	LAN (port3)	Enabled
172.17.123.0/24	172.16.123.11	DMZ (port2)	Enabled
172.27.123.0/24	172.16.123.11	DMZ (port2)	Enabled
172.16.123.0/24	172.16.123.11	DMZ (port2)	Enabled
0.0.0.0/0	Dynamic Gateway (192.168.130.1)	WAN (port1)	Enabled

Annotations and arrows explain the configuration:

- 3**: Points to the LAN (port3) interface for the first two routes, with text: "Đi vào LAN phải qua IP cổng E0/0 của Core Switch".
- 4**: Points to the DMZ (port2) interface for the next three routes, with text: "Đến DMZ (sau OpenVPN) đi qua OpenVPN".
- 2**: Points to the WAN (port1) interface for the default route, with text: "Từ bên trong ra ngoài internet".

HOẶC - Cấu hình Routing - CLI

Về vùng LAN	Đến vùng DMZ
<pre>config router static edit 1 set dst 10.101.0.0 255.255.255.0 set gateway 10.10.10.2 set device "port3" next edit 2 set dst 10.102.0.0 255.255.255.0 set gateway 10.10.10.2 set device "port3" next</pre>	<pre>config router static edit 3 set dst 172.17.123.0 255.255.255.0 set gateway 172.16.123.11 set device "port2" next edit 4 set dst 172.27.123.0 255.255.255.0 set gateway 172.16.123.11 set device "port2" next edit 5 set dst 172.16.123.0 255.255.255.0 set gateway 172.16.123.11 set device "port2" next</pre>

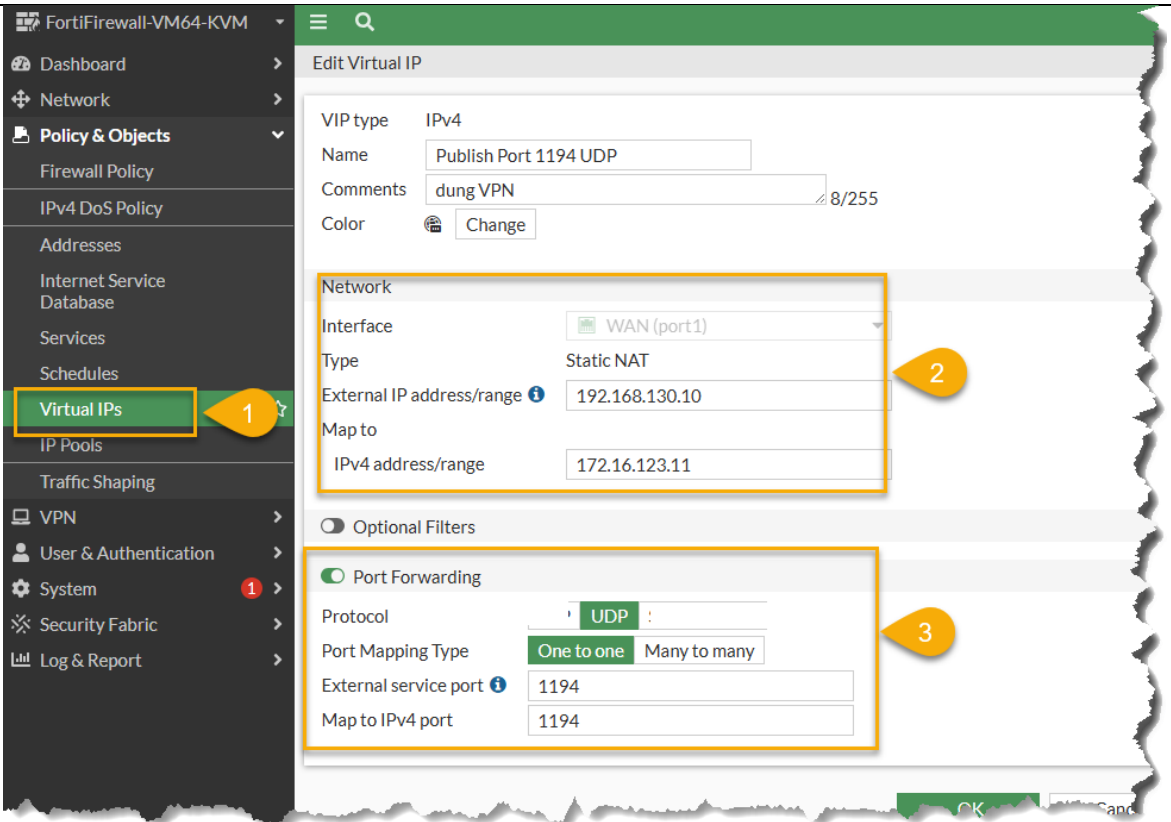
To internet	
<pre>config router static edit 6 set device "port1" set dynamic-gateway enable next end</pre>	

Cấu hình Port Forwarding/Destination NAT (DNAT)/Port Mapping/Virtual IPs (VIP) - Fortinet (FortiGate/FortiFirewall)

Ghi chú	CLI	GUI
Virtual IPs (VIP)		
Port1 WAN (public web OpenVPN)	<pre> config firewall vip edit "Public OpenVPN Web" set extip 192.168.130.10 set mappedip "172.16.123.11" set extintf "port1" set portforward enable set extport 943 set mappedport 943 next end </pre>	

Port1
WAN
(public
OpenVPN)

```
config firewall vip
edit "Publish Port 1194 UDP"
set comment "dung VPN"
set extip 192.168.130.10
set mappedip "172.16.123.11"
set extintf "port1"
set portforward enable
set protocol udp
set extport 1194
set mappedport 1194
next
end
```



Firewall Policy (chi tiết)

Ghi chú	CLI	GUI
1. Cho phép các máy từ vùng WAN có thể truy cập vào vùng DMZ		
WAN to DMZ	<pre> config firewall policy edit 2 set name "WAN-to-DMZ" set srcintf "port1" set dstintf "port2" set action accept set srcaddr "all" set dstaddr "Public OpenVPN Web" "Publish Port 1194 UDP" "Publish Port SSH" set schedule "always" set service "ALL" set logtraffic all next end </pre>	The screenshot shows the FortiGate GUI for editing a Firewall Policy named 'WAN-to-DMZ'. The configuration is as follows: Name: WAN-to-DMZ Incoming Interface: WAN (port1) Outgoing Interface: DMZ (port2) Source: all Destination: Public OpenVPN Web, Publish Port 1194 UDP, Publish Port SSH Schedule: always Service: ALL Action: ACCEPT NAT: Disabled Logging Options: Log Allowed Traffic (checked), Generate Logs when Session Starts (unchecked), Capture Packets (unchecked) Enable this policy: checked Handwritten notes in red text provide additional context: - Arrow 1 points to 'Firewall Policy' in the left sidebar. - Arrow 2 points to the 'Incoming Interface' dropdown. - Arrow 3 points to the 'Destination' list. - Arrow 4 points to the 'ACCEPT' action button. - Arrow 5 points to the 'NAT' toggle switch. - Arrow 6 points to the 'Enable this policy' toggle switch. - Note: 'Là các port đã mở, đã cấu hình trong Virtual IPs' (These are ports already open, configured in Virtual IPs) points to the destination list. - Note: 'Tắt không cần bậc' (Turn off, no need to step) points to the 'NAT' toggle.

2. DMZ to WAN/Internet

config firewall policy

edit 1

set name "DMZ-to-Internet"

set srcintf "port2"

set dstintf "port1"

set action accept

set srcaddr "all"

set dstaddr "all"

set schedule "always"

set service "ALL"

set logtraffic all

set nat enable

next

end

DMZ to
WAN

FortiGate Firewall Policy Configuration Interface

Policy Name: DMZ-to-Internet

Incoming Interface: DMZ (port2)

Outgoing Interface: WAN (port1)

Source: all

Destination: all

Schedule: always

Service: ALL

Action: ACCEPT

Firewall / Network Options:

- NAT: ☒
- IP Pool Configuration: Use Outgoing Interface Address
- Preserve Source Port: ☐

Logging Options:

- Log Allowed Traffic: ☒
- Generate Logs when Session Starts: ☐
- Capture Packets: ☐

Comments: Write a comment... 0/1023

Enable this policy: ☒

3. DMZ to LAN (nếu cần)

DMZ to
LAN

```
config firewall policy
edit 4
    set name "DMZ-to-LAN"
    set srcintf "port2"
    set dstintf "port3"
    set action accept
    set srcaddr "all"
    set dstaddr "all"
    set schedule "always"
    set service "ALL"
    set logtraffic all
next
end
```

The screenshot shows the FortiGate Firewall Policy configuration interface. The left sidebar contains a menu with 'Policy & Objects' expanded, and 'Firewall Policy' selected (marked with a yellow circle 1). The main area shows the 'Edit Policy' configuration for 'DMZ-to-LAN' (marked with a yellow circle 2). The configuration includes:

- Name: DMZ-to-LAN
- Incoming Interface: DMZ (port2)
- Outgoing Interface: LAN (port3)
- Source: all
- Destination: all
- Schedule: always
- Service: ALL
- Action: ACCEPT

Below the main configuration, the 'Firewall / Network Options' section shows 'NAT' disabled (marked with a yellow circle 3). A red dashed arrow points from the 'NAT' toggle to the text 'Tắt, Không cần NAT' (Turn off, No need NAT). The 'Logging Options' section shows 'Log Allowed Traffic' enabled, 'Generate Logs when Session Starts' disabled, and 'Capture Packets' disabled. The 'Enable this policy' toggle is also enabled (marked with a yellow circle 4). The bottom right corner has an 'OK' button.

4. LAN to WAN/Internet		
LAN to WAN	<pre>config firewall policy edit 3 set name "LAN-to-Internet" set srcintf "port3" set dstintf "port1" set action accept set srcaddr "all" set dstaddr "all" set schedule "always" set service "ALL" set logtraffic all set nat enable next end</pre>	

5. LAN to DMZ

LAN to
DMZ

```
config firewall policy
edit 5
  set name "LAN-to-DMZ"
  set srcintf "port3"
  set dstintf "port2"
  set action accept
  set srcaddr "all"
  set dstaddr "all"
  set schedule "always"
  set service "ALL"
  set logtraffic all
next
end
```

The screenshot shows the FortiGate Firewall configuration interface. The left sidebar contains the navigation menu with 'Policy & Objects' selected. The main area displays the 'Edit Policy' configuration for 'LAN-to-DMZ'. The configuration is as follows:

- Name:** LAN-to-DMZ
- Incoming Interface:** LAN (port3)
- Outgoing Interface:** DMZ (port2)
- Source:** all
- Destination:** all
- Schedule:** always
- Service:** ALL
- Action:** ACCEPT

Below the policy configuration, the 'Firewall / Network Options' section shows the 'NAT' toggle switch turned off. A red box highlights the 'NAT' toggle, and a yellow dashed arrow points from it to the text 'Tắt không cần NAT'. At the bottom, the 'Enable this policy' toggle switch is turned on, highlighted by a red box and labeled with a red circle '4'.

Firewall Policy (Tổng quan)

Nhìn tổng quan
(Các policies đi internet phải NAT)

đến Internet phải NAT

Name	Source	Destination	Schedule	Service	Action	NAT
DMZ (port2) → LAN (port3) 1						
DMZ-to-LAN	all	all	always	ALL	✓ ACCEPT	✗ Disabled
DMZ (port2) → WAN (port1) 1						
DMZ-to-Internet	all	all	always	ALL	✓ ACCEPT	✓ Enabled
LAN (port3) → DMZ (port2) 1						
LAN-to-DMZ	all	all	always	ALL	✓ ACCEPT	✗ Disabled
LAN (port3) → WAN (port1) 1						
LAN-to-Internet	all	all	always	ALL	✓ ACCEPT	✓ Enabled
WAN (port1) → DMZ (port2) 1						
WAN-to-DMZ	all	Public OpenVPN Web Publish Port 1194 UDP Publish Port SSH	always	ALL	✓ ACCEPT	✗ Disabled
Implicit 1						

❖ Switch Core

Core
1. Đặt tên cho thiết bị và tạo mới vlan
<pre>conf t hostname Core_Switch ! Vlan 99 Vlan 101-102 exit</pre>
2. Đặt địa chỉ ip cho port E0/0
<pre>Conf t interface Ethernet0/0 no switchport ip address 10.10.10.2 255.255.255.0 no shut exit</pre>
3. Đặt địa chỉ ip cho các interface vlan 99, 101, 102
<pre>interface Vlan99 ip address 10.99.0.254 255.255.255.0 no shut ! interface Vlan101 ip address 10.101.0.254 255.255.255.0 no shut ! interface Vlan102 ip address 10.102.0.254 255.255.255.0 no shut !</pre>
4. Cấu hình mode trunk cho port E1/1 và E1/2
<pre>interface Ethernet1/1 switchport trunk encapsulation dot1q switchport mode trunk exit ! interface Ethernet1/2 switchport trunk encapsulation dot1q</pre>

switchport mode trunk exit
5. Cấu hình mode Access cho port E1/3
interface Ethernet1/3 switchport access vlan 99 switchport mode access exit
6. Cấu hình Default Route trở về Firewall
ip route 0.0.0.0 0.0.0.0 10.10.10.1
7. Cấu hình DHCP Relay Agent cho các interface (trở về DHCP trong Server Farm)
interface Vlan102 ip helper-address 10.99.0.11 exit

❖ DHCP Server

1. Cấu hình căn bản	2. Cấu hình DHCP Pool
Conf t hostname DHCP_Server ! interface Ethernet0/0 ip address 10.99.0.11 255.255.255.0 exit ! ip route 0.0.0.0 0.0.0.0 10.99.0.254	ip dhcp excluded-address 10.102.0.254 ! ip dhcp pool vlan_102 network 10.102.0.0 255.255.255.0 dns-server 8.8.8.8 default-router 10.102.0.254 lease 0 0 10 end

❖ Switch Acc_01, Acc_02

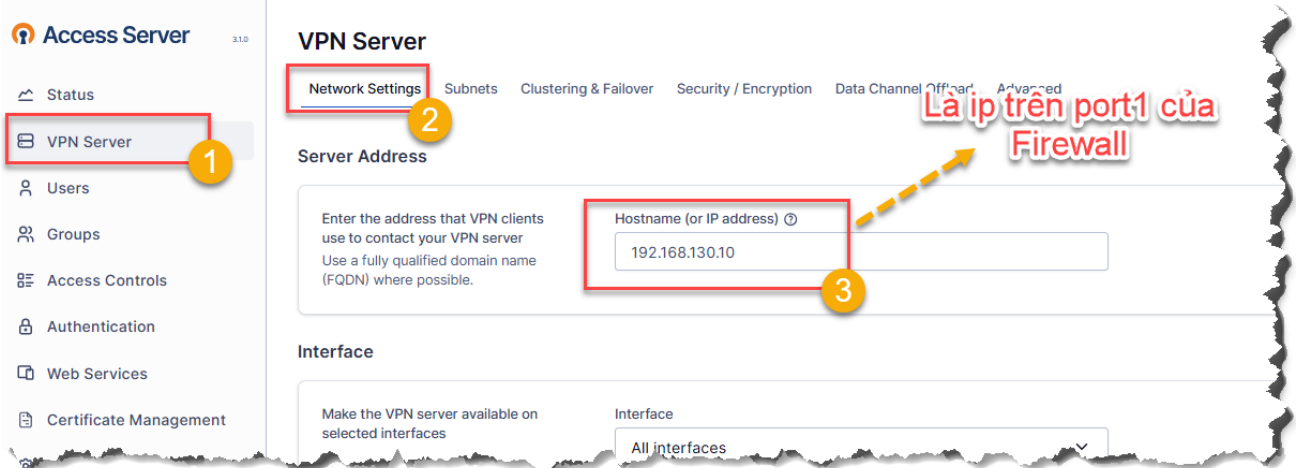
Acc_01	Acc_02
Đặt tên cho thiết bị và tạo mới vlan	
conf t hostname Acc_01 ! Vlan 101 Vlan 102	conf t hostname Acc_02 ! Vlan 102 exit

exit !	!
Cấu hình port trunk	
interface Ethernet0/0 switchport trunk encapsulation dot1q switchport mode trunk	interface Ethernet0/0 switchport trunk encapsulation dot1q switchport mode trunk
Cấu hình port access	
interface Ethernet0/1 switchport access vlan 101 switchport mode access ! interface Ethernet0/2 switchport access vlan 102 switchport mode access	interface Ethernet0/2 switchport access vlan 102 switchport mode access !

❖ vPCs

vPC_01	ip 10.101.0.11/24 10.101.0.254
vPC_02	ip 10.102.0.12/24 10.102.0.254
vPC_03	dhcp
vPC_04	dhcp

❖ OpenVPN (ID/PASS: sadmin/sadmin)

- Điền thông tin IP cho server OpenVPN, đây chính là IP hiện tại trên port1-WAN của firewall, - Khi thực hiện quay VPN thì dùng thông tin này	
- Quy hoạch/chỉ định ip cho user/nhóm cụ thể - Với cách chia này giúp việc thiết lập IP cụ thể giúp quản lý truy cập trên Firewall dễ dàng hơn	

Tạo Group/User hỗ trợ việc quản lý truy cập

Access Server 3.1.0

Status

VPN Server

Users

Groups

Access Controls

Authentication

Web Services

Certificate Management

Groups

Add New Group

Filter

Group ↑

Users in Group

Role

Auto-login

Authentication

MFA

All Users

0

User

On

local

Disabled

IT Group

1

User

On

local

Disabled

Access Server 3.1.0

Status

VPN Server

Users

Groups

Access Controls

Certificate Management

Advanced

Activation

Users

Chưa thuộc nhóm nào

Add New User

Filter

Username ↑

Role

MFA Status

Auto-login

Group

demo

User

Disabled

On

tonyteo

User

Disabled

On

IT Group

Thuộc nhóm IT Group

Vì mục đích là quản lý tập trung trên Firewall nên để ROUTE, không cần NAT, client nhận IP nào thì đến firewall bằng chính IP đó

Access Server 31.0

Access Controls

Group and User Access Rules **2**

Access Rules ⓘ

All Group User ▼ Filter

Destination ↓	Source	Protocol	Port	Reachable via
☐ 0.0.0.0/0	IT Group	All	All	Route 3

New Access Rule

Tạo Access Controls cho phép các user thuộc nhóm IT Group Route (đi ra ngoài bằng chính IP của mình nhận được)

Nếu muốn giới hạn mỗi user chỉ có **1 kết nối duy nhất** tại một thời điểm thì bạn **TẮT** nó đi

Access Server 31.0

VPN Server

Network Settings Subnets Clustering & Failover Security / Encryption Data Channel Offload **Advanced** **2**

Multiple sessions per user **3**

Enable multiple user connections ⓘ On ☐

Determines if the same user can have multiple simultaneous connections to the server.

MTU (Maximum Transmission Unit)

Set VPN tunnel MTU VPN tunnel MTU (maximum transmission unit)

Nếu on Hỗ trợ nhiều thiết bị có thể dùng chung 1 user để kết nối

9. TEST VPN Client to Site

❖ Trên OpenVPN Server (Access Server)

Máy server chọn user cần dùng VPN, tải profile về sau đó chuyển đến máy client

Access Server 3.1.0

Users

Filter

Username ↑ Role MFA Status Auto-login Group

demo User Disabled On

tonyteo User Disabled On

Edit

Block

Delete

Change group

Reset Password

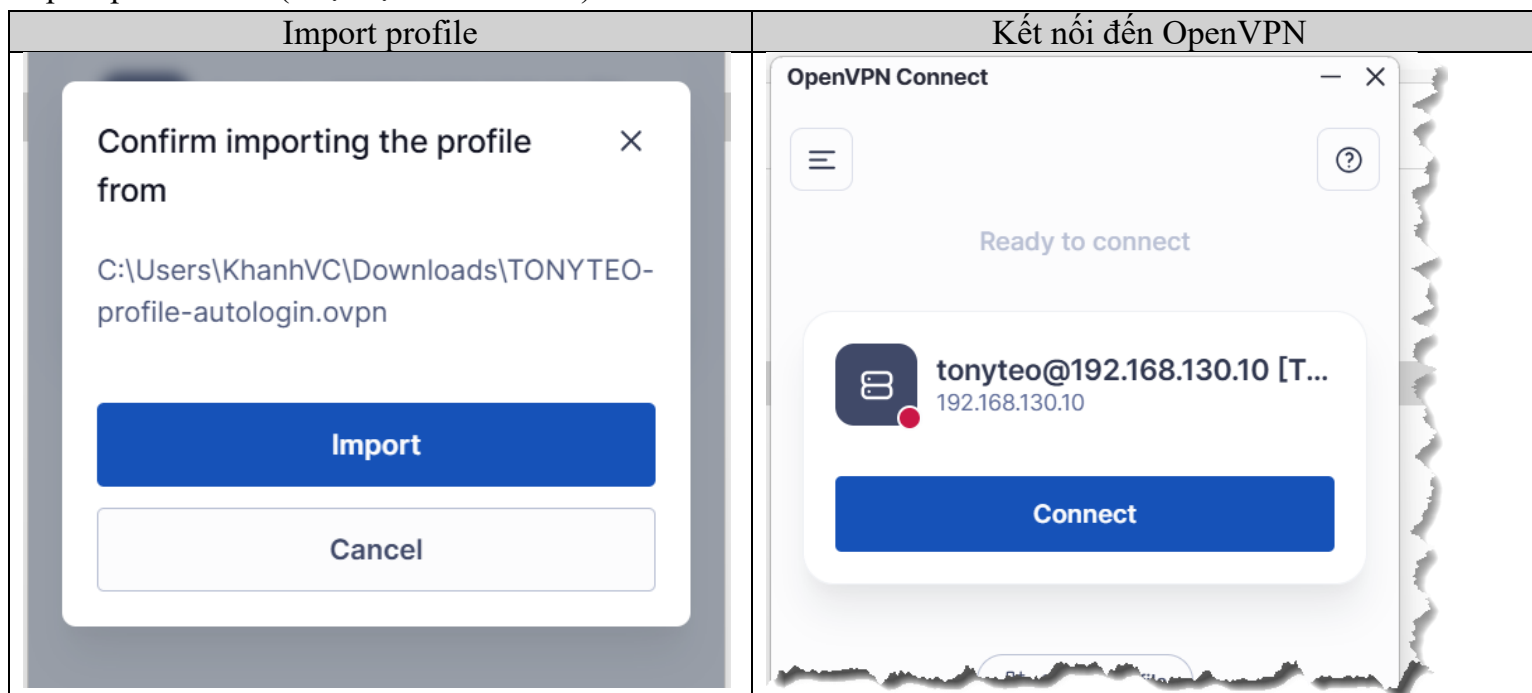
Download profile

Enable MFA

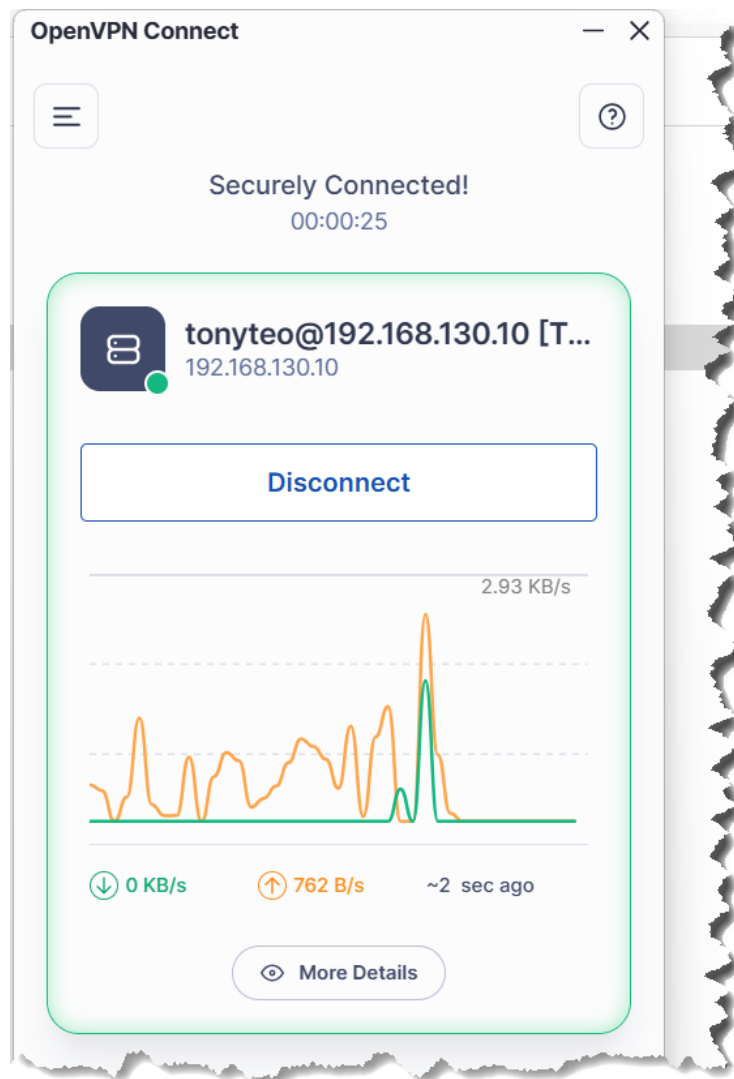
Add New User

❖ Cài OpenVPN trên máy con

- Tải openvp về cài đặt <https://openvpn.net/client/>
- Import profile user (được tải từ bước trên) và kết nối



Kết quả



Video theo tham khảo

How to Import Labs in EVE NG

<https://youtu.be/YHVJWJVoG3U?si=C6JioUlsRwtjzlj9>